COUNTABLE INDEXEDNESS ASCENDS AND DESCENDS ALONG A REPRESENTABLE SURJECTIVE FLAT MORPHISM OF AFFINE FORMAL ALGEBRAIC SPACES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof displays a commutative diagram with a dotted arrow, whose existence is the technical heart of the argument; it lies inside the counted span and is reproduced verbatim.

Editorial note. The counted spans carry 6 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Presentations as filtered colimits, and factorisation of morphisms from quasi-compact objects through a stage. Same chapter, Lemma lemma-countable-affine-formal-algebraic-space at proof lines 4555-4556, Definition definition-affine-formal-algebraic-space at proof lines 4557-4558, and Lemma lemma-factor-through-thickening at proof lines 4560-4562 and 4565-4567. Supply the two presentations and the two factorisation steps.
- The fibre product of an affine scheme with the source over the target is affine. Same chapter, Lemma lemma-affine-representable-by-algebraic-spaces, invoked at proof lines 4562-4564. Makes the fibre products appearing in the argument affine schemes.
- An inclusion of ideals can be tested after a faithfully flat extension. Standard, used at proof lines 4584-4589. Gives the existence of the dotted arrow from the containment after extending along the source morphism.
- Countable indexedness descends along a representable surjective flat morphism of affine formal algebraic spaces. Same chapter, Lemma lemma-property-goes-up-affine, invoked at proof lines 4591-4593. Gives the converse direction.

Source reference. Original lemma 0AKN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4209–4209. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AII: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2423–2423. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIA: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2080–2080. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AKQ: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 4373–4373. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
   (a) each X_λ is affine,
   (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

such that

$$X \cong \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let $X \to Y$ be a morphism of affine formal algebraic spaces which is representable by algebraic spaces, surjective, and flat. Then X is countably indexed if and only if Y is countably indexed.*

Proof. Assume X is countably indexed. We write $X = \operatorname{colim} X_n$ as in Lemma 0AII. Write $Y = \operatorname{colim} Y_\lambda$ as in Definition 0.1. For every n we can pick a λ_n such that $X_n \to Y$ factors through Y_{λ_n}, see Lemma 0AIA. On the other hand, for every λ the scheme $Y_\lambda \times_Y X$ is affine (Lemma 0AKN) and hence $Y_\lambda \times_Y X \to X$ factors through X_n for some n (Lemma 0AIA). Picture

$$\begin{array}{ccccc} Y_\lambda \times_Y X & \longrightarrow & X_n & \longrightarrow & X \\ \downarrow & & \downarrow & & \downarrow \\ Y_\lambda & \dashrightarrow & Y_{\lambda_n} & \longrightarrow & Y \end{array}$$

If we can show the dotted arrow exists, then we conclude that $Y = \operatorname{colim} Y_{\lambda_n}$ and Y is countably indexed. To do this we pick a μ with $\mu \geq \lambda$ and $\mu \geq \lambda_n$. Thus both $Y_\lambda \to Y$ and $Y_{\lambda_n} \to Y$ factor through $Y_\mu \to Y$. Say $Y_\mu = \operatorname{Spec}(B_\mu)$, the closed subscheme Y_λ corresponds to $J \subset B_\mu$, and the closed subscheme Y_{λ_n} corresponds to $J' \subset B_\mu$. We are trying to show that $J' \subset J$. By the diagram above we know $J'A_\mu \subset JA_\mu$ where $Y_\mu \times_Y X = \operatorname{Spec}(A_\mu)$. Since $X \to Y$ is surjective and flat the morphism $Y_\lambda \times_Y X \to Y_\lambda$ is a faithfully flat morphism of affine schemes, hence $B_\mu \to A_\mu$ is faithfully flat. Thus $J' \subset J$ as desired.

Assume Y is countably indexed. Then X is countably indexed by Lemma 0AKQ.
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